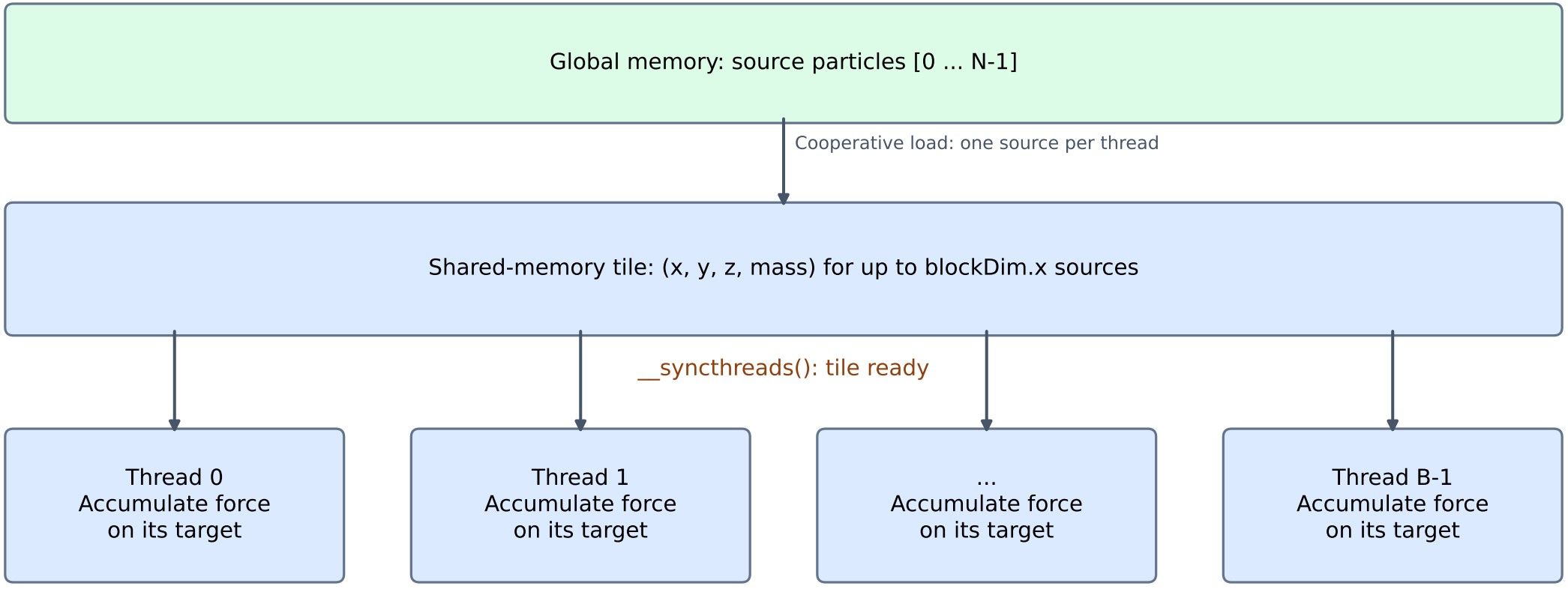
CUDA: reuse a particle tile inside each block

One thread owns one target particle. Source particles are processed in increasing index order.



Barrier before replacing the tile; repeat for all source tiles, then update velocities.

A separate kernel updates positions. Naive baseline reads sources directly from global memory without a shared tile.